SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I S . Sahil -192525120 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

S . Sahil

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Grey level transformation is an image enhancement technique used to modify the intensity values of pixels. It improves the visual quality of an image by enhancing brightness, contrast, and highlighting important details, making the image easier to analyse.

(b) Apply a negative transformation s=255-r to the pixel values: r=[50,100,150,200]
Compute the transformed values.

Formula: s = 255-r

Original	Transformed(s)
50	205
100	155
150	105
200	55

Answer: 205, 155, 105, 55

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Log transformation enhances low-intensity (dark) pixels by increasing their brightness while compressing high-intensity (bright) pixels. This helps reveal hidden details in dark regions of an image without significantly affecting brighter areas.

(b) Given $s = c\log(1+r)$, where $c=10$, compute s for: $r=[0,10,50,100]$

Formula: $s=10\log(1+r)$

Original (r)	Calculation	Transformed (s)
0	10 log(1)	0.00
10	10 log(11)	23.98
50	10 log(51)	39.32
100	10 log(101)	46.15

Answer: 0.00, 23.98, 39.32, 46.15

3. Histogram Processing
(a) Explain the role of histogram equalization in contrast enhancement.

Histogram equalization is used to improve the contrast of an image by redistributing the intensity values more evenly across the available range. This makes dark areas brighter, bright areas clearer, and improves the overall visibility of image details.

(b) Given a 3-bit image with histogram:

Intensity	**Frequency**
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Given Histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total Pixels = 16

Cumulative Distribution and Equalized Values:

Intensity	Frequency	Cumulative Frequency	CDF	Equalized Intensity
0	2	2	0.125	1
1	2	4	0.250	2
2	4	8	0.500	4
3	8	16	1.000	7

Answer: Equalized intensity values = 1, 2, 4, 7

4. Spatial Filtering (Smoothing)
(a) Explain the purpose of spatial filtering for smoothing.

Spatial filtering for smoothing is used to reduce noise and unwanted variations in an image. It replaces each pixel with the average of its neighbouring pixels, producing a smoother image while reducing small intensity fluctuations.

(b) Apply a 3×3 averaging filter to the following region:
10 20 30 20 30 40 30 40 50
Compute the new center pixel value.

Input Matrix:

10	20	30
20	30	40
30	40	50

Calculation:

$$\frac{10+20+30+20+30+40+30+40+50}{9} = 270 / 9 = 30$$

Answer: New centre pixel value = 30

5. Spatial Filtering (Sharpening)
(a) Explain how sharpening enhances image details.

Sharpening improves image quality by enhancing edges and fine details. It increases the contrast between neighbouring pixels, making object boundaries and textures appear clearer and more distinct.

(b) Apply the Laplacian mask:
[0 -1 0}, {-1 4 -1], { 0 -1 0]
to the matrix:
10 10 10, 10 20 10, 10 10 10
Compute the output at the center.

Laplacian Mask:

0	-1	0
-1	4	-1
0	-1	0

Input Matrix:

10	10	10
10	20	10
10	10	10

Calculation:

$$(4\times20) - (10+10+10+10)=80-40 = 40$$

Answer: Output at the centre pixel = 40